

High Performance Computing

ASSIGNMENT-8

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Assignment 1: Serial Performance Baseline and Profiling (Amdahl's Law)

Objective: Measure speedup for different thread counts and identify serial regions.

Python Code

```
import time
from concurrent.futures import ProcessPoolExecutor

# Simulate a compute-heavy task (Parallel Region)
def compute_chunk(start, end):
    total = 0
    for i in range(start, end):
        total += i * i
    return total

def run_assignment_1(total_work, num_threads):
    # 1. Serial Region (cannot be parallelized)
    start_serial = time.time()
    time.sleep(0.2) # Simulating I/O or setup
    serial_time = time.time() - start_serial

    # 2. Parallel Region
    start_parallel = time.time()
    chunk_size = total_work // num_threads

    futures = []
    with ProcessPoolExecutor(max_workers=num_threads) as executor:
        for i in range(num_threads):
            start = i * chunk_size
            end = total_work if i == num_threads - 1 else (i + 1) * chunk_size
            futures.append(executor.submit(compute_chunk, start, end))

    # Wait for all futures to complete
    for future in futures:
        future.result()
```

```

# Wait for all to finish results = [f.result()
for f in futures] parallel_time =
time.time() - start_parallel

total_time = serial_time + parallel_time
return total_time

if __name__ ==
'__main__': # USER
INPUT total_work =
50_000_000
print(f"--- Assignment 1: Amdahl's Law Profiling ---")
print(f"Total Workload: {total_work} iterations\n")

# Baseline (1 thread)
base_time = run_assignment_1(total_work, 1)
print(f"Threads: 1 | Time: {base_time:.4f}s | Speedup: 1.00x | Efficiency: 100.0%")

for threads in [2, 4, 8]: exec_time =
    run_assignment_1(total_work, threads)
    speedup = base_time / exec_time efficiency = (speedup / threads) * 100
print(f"Threads: {threads} | Time: {exec_time:.4f}s | Speedup: {speedup:.2f}x | Efficiency:
{efficiency:.1f}%")

```

Output

User Input (Parameters): Total Workload = 50,000,000

Output:

--- Assignment 1: Amdahl's Law Profiling ---

Total Workload: 50000000 iterations

Threads: 1 | Time: 2.3412s | Speedup: 1.00x | Efficiency: 100.0%

Threads: 2 | Time: 1.3021s | Speedup: 1.80x | Efficiency: 90.0%

Threads: 4 | Time: 0.7845s | Speedup: 2.98x | Efficiency: 74.5%

Threads: 8 | Time: 0.5213s | Speedup: 4.49x | Efficiency: 56.1%

Assignment 2: Hotspot Detection Using Timing Analysis

Objective: Insert timers around code sections to detect the most time-consuming regions.

Python Code

```
import time
import math

def do_initialization(size):
    return [0] * size

def do_computation(data):
    # Hotspot region
    return [math.sqrt(i) ** 2.5 for i in range(len(data))]

def do_io(data):
    # Simulating writing to
    # disk time.sleep(0.1)
    return True

if __name__ == '__main__':
    # USER INPUT
    array_size = 5_000_000
    print(f"--- Assignment 2: Hotspot Detection ---")

    # 1. Measure Initialization
    t0 = time.time()
    data = do_initialization(array_size)
    t1 = time.time()
    init_time = t1 - t0

    # 2. Measure Computation t0
    = time.time() result =
    do_computation(data) t1 =
    time.time() comp_time = t1 -
    t0

    # 3. Measure
    I/O t0 =
    time.time()
    do_io(result) t1
    = time.time()
    io_time = t1 - t0
```

```

print(f"Initialization Time: {init_time:.4f} seconds")
print(f"Computation Time: {comp_time:.4f} seconds")
print(f"I/O Time: {io_time:.4f} seconds")

times = {"Initialization": init_time, "Computation": comp_time, "I/O": io_time}
hotspot = max(times, key=times.get)
print(f"\n=> HOTSPOT IDENTIFIED: '{hotspot}' consumes the maximum runtime.")
print("=> Action: Focus parallelization efforts on this region.")

```

Output

User Input (Parameters): Array Size = 5,000,000

Output:

```

--- Assignment 2: Hotspot Detection ---
Initialization Time: 0.0150 seconds
Computation Time: 0.8920 seconds
I/O Time: 0.1012 seconds

```

Assignment 3: Load Imbalance Detection

Objective: Detect workload imbalance and improve distribution using different scheduling strategies (Static vs. Dynamic).

Python Code

```

import time
from multiprocessing import Pool

# Simulating uneven workloads (e.g., some loops take much longer)
def uneven_task(task_id):
    sleep_time = 0.5 if task_id % 4 == 0 else 0.05
    time.sleep(sleep_time)
    return task_id

if __name__ == '__main__':
    # USER INPUT

```

```

tasks = list(range(16)) # 16 tasks with uneven weights

num_threads = 4 print(f"--- Assignment 3: Load

Imbalance Detection ---")

# 1. Static Scheduling (equivalent to OpenMP schedule(static))
# Chunks are assigned sequentially, leading to imbalance if heavy tasks cluster
start_time = time.time() with Pool(processes=num_threads) as pool: #
chunksize=4 forces rigid static allocation
    results_static = pool.map(uneven_task, tasks, chunksize=4)
static_time = time.time() - start_time
print(f"Execution time with STATIC scheduling: {static_time:.4f}s")

# 2. Dynamic Scheduling (equivalent to OpenMP schedule(dynamic))
# Workers pick up tasks as they become free
start_time = time.time() with
Pool(processes=num_threads) as pool: #
imap_unordered with chunksize=1 acts dynamically
    results_dynamic = list(pool.imap_unordered(uneven_task, tasks, chunksize=1))
dynamic_time = time.time() - start_time
print(f"Execution time with DYNAMIC scheduling: {dynamic_time:.4f}s")

print("\nConclusion: Dynamic scheduling improves performance by balancing the uneven workload
among available threads.")

```

Output

User Input (Parameters): 16 Tasks, 4 Threads. **Output:**

```

--- Assignment 3: Load Imbalance Detection --Execution
time with STATIC scheduling: 1.6214s
Execution time with DYNAMIC scheduling: 0.7105s

```

Assignment 4: False Sharing and Memory Bottlenecks

Objective: Identify memory-related performance issues (simulating false sharing via lock contention vs private memory).

Python Code

```
import time
```

```
from multiprocessing import Process, Array, Value
```

```
# Scenario A: Contention/False Sharing on a shared data
structure def bad_parallel_update(shared_array, index,
iterations): for _ in range(iterations):
    # Continuous hammering of shared memory space
    shared_array[index] += 1
```

```
# Scenario B: Private arrays (padding equivalent) def
good_parallel_update(shared_array, index, iterations):
    local_sum = 0
    # Process locally in private memory/cache
    for _ in range(iterations): local_sum += 1
    # Write back to shared memory ONLY ONCE at the end
    shared_array[index] = local_sum
```

```
if __name__ == '__main__': # USER INPUT num_threads = 4
iterations = 2_000_000 print(f"--- Assignment 4: False Sharing &
Memory Bottlenecks ---")
```

```
shared_arr_bad = Array('i', num_threads)
shared_arr_good = Array('i', num_threads)
```

```
# 1. Measure Performance Degradation (Shared contention)
t0 = time.time()
processes = [Process(target=bad_parallel_update, args=(shared_arr_bad, i, iterations)) for i in
range(num_threads)]
for p in processes: p.start()
for p in processes: p.join()
t_bad = time.time() - t0
```

```
# 2. Modify Data Layout (Private memory/Padding)
t0 = time.time()
processes = [Process(target=good_parallel_update, args=(shared_arr_good, i, iterations)) for i in
range(num_threads)]
for p in processes: p.start()
for p in processes: p.join()
t_good = time.time() - t0
```

```
print(f"Time with Memory Contention (Simulated False Sharing):  
{t_bad:.4f}s") print(f"Time with Private Memory Layout (Optimized):  
{t_good:.4f}s") print(f"Improvement Factor: {t_bad / t_good:.2f}x faster")
```

Example User Input & Output

User Input (Parameters): 4 Threads, 2,000,000 iterations.

Expected Output:

```
--- Assignment 4: False Sharing & Memory Bottlenecks --Time  
with Memory Contention (Simulated False Sharing): 4.1520s  
Time with Private Memory Layout (Optimized): 0.0812s Improvement  
Factor: 51.13x faster
```

Assignment 5: Synchronization Overhead Analysis

Objective: Measure synchronization overhead and replace critical sections with reductions or atomics.

Python Code

```
import time  
from multiprocessing import Process, Value, Lock  
  
# Scenario A: Using a Critical Section (Lock) for every operation  
def critical_section_sum(shared_sum, lock, iterations):  
    for i in range(iterations):  
        with lock: # Synchronization overhead  
            shared_sum.value += 1  
  
# Scenario B: Using Reduction (Compute locally, synchronize once) def  
reduction_sum(shared_sum, lock, iterations):  
    local_sum = 0  
    for i in range(iterations):  
        local_sum += 1  
  
# Synchronization happens ONLY ONCE at the end  
with lock: shared_sum.value +=  
    local_sum
```

```

if __name__ == '__main__': # USER INPUT num_threads = 4
    iterations = 500_000 print(f"--- Assignment 5: Synchronization
    Overhead Analysis ---") lock = Lock()

    # 1. Heavy Synchronization (Critical)
    shared_val_critical = Value('i', 0) t0 =
    = time.time()
    processes = [Process(target=critical_section_sum, args=(shared_val_critical, lock, iterations)) for _ in
    range(num_threads)]
    for p in processes: p.start()
    for p in processes: p.join()
    t_critical = time.time() - t0

    # 2. Reduced Synchronization (Reduction)
    shared_val_reduction = Value('i', 0) t0 =
    time.time()
    processes = [Process(target=reduction_sum, args=(shared_val_reduction, lock, iterations)) for _ in
    range(num_threads)]
    for p in processes: p.start()
    for p in processes: p.join()
    t_reduction = time.time() -
    t0

    print(f"Execution time with Excessive Critical Sections: {t_critical:.4f}s")
    print(f"Execution time with Local Reduction: {t_reduction:.4f}s")
    print(f"\nOptimization Result: Eliminated unnecessary barriers and critical sections, improving speed
    significantly.")

```

Output

User Input (Parameters): 4 Threads, 500,000 iterations. **Output:**

```

--- Assignment 5: Synchronization Overhead Analysis ---
Execution time with Excessive Critical Sections: 3.8201s Execution
time with Local Reduction:      0.0450s

```